A Diophantine and Structural Study of the Erdős–Selfridge Prime-Power Variant of Problem #889

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Abstract

We study the prime-power variant introduced by Erdős and Selfridge in their 1967 paper on prime factors of consecutive integers. For

$$V(n,k) = \#\{p : \ p^{\nu_p(n+k)} > k\}, \qquad V_1(n) = \max_{k\geq 1} V(n,k),$$

Erdős and Selfridge asked whether the set

$$\{n \geq 1 : \ V_1(n) = 1\}$$

is finite; they identified

$$1, 2, 3, 4, 6, 7, 8, 12, 15, 16, 24, 30, 48, 80$$

as the known values below 2500, and remarked that 80 was probably the largest solution. The question remains open.

The purpose of this paper is not to claim a resolution. Instead, we develop a detailed Diophantine reduction of the problem and record the advances obtained so far. We first reformulate the condition in terms of the ordered prime-power components of $n + k$, and derive the decomposition

$$n + k = d_k q_k, \qquad d_k \mid \operatorname{lcm}(1, \ldots, k),$$

together with the decisive compatibility condition

$$\gcd(d_i, d_j) \mid |i - j|.$$

We then give a complete proof that the branch $d_2 = 1$ produces only the seven small values

$$1, 2, 3, 6, 7, 15, 30,$$

using Catalan–Mihăilescu, Bang–Zsigmondy, and elementary congruences. For the branch $d_2 = 2, d_3 = 1$, we prove that if the large prime-power component of $n + 3$ has exponent > 1, then the branch collapses to the single exception $n = 24$. We then analyze the complementary case in which $n + 3$ is prime. If

$$n + 1 = P^{\alpha}, \qquad n + 2 = 2Q^{\beta},$$

the exponents are sharply constrained: when $\alpha > 1$, only the two known values $n = 8, 80$ remain, while when $\alpha = 1$ and $\beta > 1$ the problem reduces to the residual simultaneous-primality family

$$2 \cdot 3^{2^s} - 1, \qquad \frac{3^{2^s} + 1}{2}, \qquad 2 \cdot 3^{2^s} + 1.$$

We deliberately leave that final family open rather than claiming a resolution not supplied by the present arguments.

A further elementary sieve using the primes $7, 11, 19, 47, 89$, and 311 shows that its parameter must satisfy

$$s \equiv 5 \text{ or } 9 \pmod{12}, \qquad s \not\equiv 9 \pmod{11}, \qquad s \not\equiv 9 \pmod{10}, \qquad s \not\equiv 5 \pmod{20}.$$

In particular, none of the residual parameters $2 \le s \le 16$ survives. Thus any counterexample arising from this residual family must have

$$n \ge 2 \cdot 3^{131072} - 2.$$

The remaining leaves are also reduced explicitly. In particular, for $d_2 = 2, d_3 = 3$ we obtain $12 \mid n$, the dichotomy $d_4 \in \{1, 4\}$, the collapse $d_4 = 1 \Rightarrow n = 12$, and finally $d_5 \in \{1, 5\}$. The subbranch $d_5 = 1$ yields a system of four consecutive prime-power equations, while $d_5 = 5$ yields a chain of linear relations between prime powers and, at $k = 6$, an explicit exponential family when $d_6 = 2$.

An exact finite computation up to $5 \cdot 10^7$, recorded together with the research code, finds no new exceptions and exhibits a very rigid hierarchy of minimal witnesses. We emphasize throughout the distinction between proved statements, named-theorem consequences, and computational observations.

1 Research status and scope

The original Erdős–Selfridge Problem #889 asks whether

$$v_0(n) := \max_{k \ge 0} v(n, k) \longrightarrow \infty \qquad (n \to \infty),$$

where $v(n, k)$ counts distinct prime factors of $n + k$ exceeding k. The problem is currently listed as open, and the formalized statement in the Google DeepMind *Formal Conjectures* project is likewise marked `research open`. The prime-power variant discussed in this paper is also explicitly represented there as an open statement. See [1, 2, 3].

Accordingly, the central objective of this manuscript is to build a rigorous partial theory rather than to advertise a complete solution. In particular, all branches that are not actually proved to be impossible are kept visible in the final decision tree.

There are three different levels of information used below.

(i) **Elementary or named-theorem proofs.** These are lemmas proved in the paper, sometimes using standard results such as Catalan–Mihăilescu or Bang–Zsigmondy.

(ii) **Exact computational observations.** These concern only a finite interval and are not used as substitutes for an infinite proof.

(iii) **Research targets.** These are surviving Diophantine systems for which we do not yet have a complete elimination.

This distinction is essential: the problem remains open even though a large finite range is completely understood computationally.

2 The Erdős–Selfridge prime-power variant

2.1 Prime-power components

For an integer $m \ge 2$, write

$$m = \prod_{p \mid m} p^{\nu_p(m)}.$$

For each prime divisor $p \mid m$, define its full prime-power component by

$$Q_p(m) := p^{\nu_p(m)}.$$

Order the distinct components decreasingly:

$$Q_1(m) > Q_2(m) > \cdots > Q_{\omega(m)}(m).$$

If m has only one prime divisor, put $Q_2(m) = 0$.

We study

$$V(n,k) = \#\{p : \ Q_p(n+k) > k\} = \#\{p : \ p^{\nu_p(n+k)} > k\},$$

and

$$V_1(n) = \max_{k \geq 1} V(n,k).$$

Thus $V(n,k) \leq 1$ means that among the full prime-power components of $n+k$, at most one exceeds k.

Lemma 2.1. *For every $n \geq 1$ and $k \geq 1$,*

$$V(n,k) \geq 2 \iff Q_2(n+k) > k.$$

Proof. By definition, $V(n,k) \geq 2$ exactly when at least two distinct prime-power components of $n+k$ exceed k. This is equivalent to the second largest such component exceeding k. □

2.2 Interval formulation

Writing $m = n+k$, the inequality $Q_2(m) > k$ becomes

$$m - n > 0, \qquad m - n < Q_2(m).$$

Equivalently,

$$m - Q_2(m) + 1 \leq n \leq m - 1.$$

Hence every integer m contributes an interval

$$\boxed{m - Q_2(m) + 1 \leq n \leq m - 1} \tag{1}$$

on which $V(n, m-n) \geq 2$.

Consequently, proving that $V_1(n) \geq 2$ for all sufficiently large n is equivalent to proving that these intervals cover all sufficiently large integers.

2.3 A universal bound on the offset

Lemma 2.2. *If $V(n,k) \geq 2$, then*

$$k^2 - k < n.$$

Proof. If $V(n,k) \geq 2$, there are two distinct prime-power components $A, B > k$ dividing $n+k$. Since they correspond to distinct primes, they are coprime, so

$$AB \mid n+k.$$

Therefore

$$n + k \geq AB > k^2,$$

which gives

$$n > k^2 - k.$$

□

Thus for a finite computation $n \leq N$, it suffices to consider

$$1 \leq k \leq K(N), \qquad K(N) = \max\{k : \ k^2 - k < N\}.$$

3 Exact finite computation

The accompanying exact sieve computes the second-largest prime-power component $Q_2(m)$ for every

$$m \le N + K(N)$$

using a smallest-prime-factor table. The interval representation (1) is then accumulated with a difference array.

For

$$N = 50\,000\,000, \qquad K(N) = 7071,$$

the reported exact computation finds precisely

$$\boxed{\mathcal{E} = \{1, 2, 3, 4, 6, 7, 8, 12, 15, 16, 24, 30, 48, 80\}} \tag{2}$$

as the values with $V_1(n) = 1$.

This is a *finite computational theorem*: it asserts the complete answer inside the stated range and nothing about larger n.

The same computation defines the minimal witness

$$\kappa(n) := \min\{k \ge 1 : \ V(n,k) \ge 2\}$$

for every non-exceptional $n \le 50\,000\,000$. The recorded distribution is

$\kappa(n)$	number of n
1	46 997 805
2	2 876 654
3	113 669
4	11 288
5	484
6	80
7	4
8	2

(3)

and the last six integers surviving the tests $k \le 6$ are

$$240, \quad 471240, \quad 5516280, \quad 16831080, \quad 18164160, \quad 29743560. \tag{4}$$

Four are eliminated at $k = 7$, while

$$\kappa(5516280) = \kappa(18164160) = 8.$$

An important warning is worth recording. The number 240 is *not* an exception to the prime-power variant: indeed

$$240 + 7 = 247 = 13 \cdot 19,$$

so $V(240, 7) = 2$. It is only an exceptionally late finite-range survivor of the tests $k \le 6$.

4 The divisor decomposition

From now on, assume

$$V_1(n) = 1.$$

For each k, put

$$L_k := \operatorname{lcm}(1, 2, \ldots, k).$$

All but at most one prime-power component of $n+k$ are $\le k$. Their product therefore divides L_k. Hence we may write

$$\boxed{n+k = d_k q_k, \qquad d_k \mid L_k, \qquad \gcd(d_k, q_k) = 1,} \tag{5}$$

where q_k is either 1 or the unique prime-power component which may exceed k.

For large n, once $k \ge 8$, $q_k = 1$ is impossible because $n + k > L_k$ for the relevant range.

Lemma 4.1. *For $i \neq j$,*

$$\boxed{\gcd(d_i, d_j) \mid |i-j|.} \tag{6}$$

In particular,

$$\boxed{\gcd(d_k, d_{k+1}) = 1.} \tag{7}$$

Proof. Suppose a prime ℓ divides both d_i and d_j. Then $\ell \mid n+i$ and $\ell \mid n+j$, hence

$$\ell \mid (n+i) - (n+j) = i - j.$$

The same statement prime-by-prime gives the divisibility claim for the gcd. The consecutive case follows because $|k - (k+1)| = 1$. □

The simple relation (6) is the basic mechanism behind the Diophantine decision tree.

5 Complete treatment of the branch $d_2 = 1$

We now prove a substantial theorem which closes the entire $d_2 = 1$ branch.

5.1 Two consecutive prime powers

Since $d_1 = 1$, we have

$$n + 1 = p^a.$$

If $d_2 = 1$, then

$$n + 2 = q^b.$$

Thus $n+1$ and $n+2$ are consecutive prime powers.

One of the two numbers is even, and therefore one of the underlying primes is 2.

5.1.1 Case I: $n + 1 = 2^a$

Then

$$n + 2 = 2^a + 1 = q^b.$$

If $a, b > 1$, Catalan–Mihăilescu gives the unique pair of consecutive perfect powers $8, 9$, hence

$$n = 7.$$

If $a = 1$, then $n = 1$.

Suppose $b = 1$. Then

$$2^a + 1$$

is prime. For $a > 1$, a must be a power of 2: if $a = rs$ with $r > 1$ odd, then

$$2^a + 1 = (2^s)^r + 1$$

is composite. Hence

$$a = 2^t, \qquad t \ge 1.$$

The values $t = 1, 2$ give

$$n = 3, \qquad n = 15.$$

It remains to eliminate

$$\boxed{n = 2^{2^t} - 1, \qquad t \geq 3.} \tag{8}$$

Lemma 5.1. *For every $t \geq 3$,*

$$n = 2^{2^t} - 1$$

satisfies $V(n, k) \geq 2$ for some $k \leq 4$.

Proof. Put

$$M = 2^t.$$

Suppose first that t is even. Then $t \geq 4$, and

$$A := M - 1 = 2^t - 1$$

is an odd composite integer.

Now

$$n + 3 = 2^M + 2 = 2(2^A + 1).$$

Choose a proper divisor $u \mid A$ with $u \geq 5$. Such a divisor exists because $A \geq 15$ is odd composite. Then

$$2^u + 1 \mid 2^A + 1.$$

By the Bang–Zsigmondy theorem there exists a primitive prime divisor r of

$$2^{2A} - 1,$$

so

$$\mathrm{ord}_r(2) = 2A.$$

Likewise, since $u \geq 5$, there exists a primitive prime divisor ℓ of

$$2^{2u} - 1,$$

with

$$\mathrm{ord}_\ell(2) = 2u.$$

Both r and ℓ divide $2^A + 1$, while their orders are different, so $r \neq \ell$. Moreover both primes exceed 3. Thus $n + 3$ has at least two prime-power components > 3, and

$$V(n, 3) \geq 2.$$

Now suppose t is odd. Then

$$M = 2^t \equiv 2 \pmod 3.$$

Since 2 has order 3 modulo 7,

$$2^M + 3 \equiv 2^2 + 3 \equiv 4 + 3 \equiv 0 \pmod 7.$$

Hence

$$7 \mid n + 4.$$

Also $n + 4$ is odd and

$$n + 4 = 2^M + 3 \equiv 1 \pmod 3,$$

so the quotient $(n + 4)/7$ is an integer > 1 with no prime factor 2 or 3. Hence it has a prime divisor > 3, distinct from 7. Consequently

$$V(n, 4) \geq 2.$$

□

Thus the family (8) contains no counterexample.

5.1.2 Case II: $n + 2 = 2^b$

Now

$$n + 1 = p^a, \qquad n + 2 = 2^b.$$

If $a, b > 1$, Catalan–Mihăilescu would force the pair $8, 9$, but the larger member here would have to be 2^b, impossible. Hence

$$a = 1.$$

Therefore

$$n = 2^b - 2, \qquad p = 2^b - 1.$$

Since p is prime,

$$\boxed{b \text{ is prime.}} \tag{9}$$

The small primes give

$$b = 2, 3, 5 \quad \Longrightarrow \quad n = 2, 6, 30.$$

Suppose $b > 3$.
If

$$b \equiv 3 \pmod 4,$$

then

$$n + 4 = 2^b + 2 = 2(2^{b-1} + 1).$$

Since $b - 1 \equiv 2 \pmod 4$,

$$5 \mid 2^{b-1} + 1,$$

while $3 \nmid 2^{b-1} + 1$. Since $V(n, 4) \leq 1$, every prime factor of the odd number $2^{b-1} + 1$ other than possibly 5 would be another prime > 4. Hence

$$2^{b-1} + 1 = 5^c.$$

Modulo 8, this gives c even. Thus

$$5^c - 2^{b-1} = 1$$

is a pair of consecutive perfect powers with both exponents > 1, which contradicts Catalan–Mihăilescu. Therefore

$$\boxed{b \equiv 1 \pmod 4.} \tag{10}$$

We sharpen this.

Lemma 5.2. *Under $d_2 = 1$, if $n = 2^b - 2$ with $b > 5$, then*

$$b = 2^{2^s} + 1$$

for some $s \geq 2$.

Proof. We have

$$n + 4 = 2(2^{b-1} + 1).$$

Put

$$A = b - 1.$$

By (10), $4 \mid A$.
Suppose A has an odd divisor $r > 1$. Write

$$A = rs.$$

Then s is also divisible by 4, and

$$2^s + 1 \mid 2^A + 1.$$

A primitive divisor ℓ of

$$2^{2s} - 1$$

has

$$\mathrm{ord}_\ell(2) = 2s.$$

A primitive divisor R of

$$2^{2A} - 1$$

has

$$\mathrm{ord}_R(2) = 2A.$$

Both divide $2^A + 1$, and the orders are different, so $\ell \neq R$. Since $s \geq 4$, both primes exceed 4, contradicting $V(n, 4) \leq 1$.

Hence A has no odd divisor and therefore

$$A = 2^t.$$

Thus

$$b = 2^t + 1.$$

Since b is prime, t itself must be a power of 2. Therefore

$$b = 2^{2^s} + 1.$$

□

It remains to eliminate these Fermat-prime-shaped values.

Lemma 5.3. *Let*

$$b = 2^{2^s} + 1, \qquad s \geq 2,$$

and

$$n = 2^b - 2.$$

Then

$$V(n, 6) \geq 2.$$

Proof. Put

$$M = b - 2 = 2^{2^s} - 1.$$

Since $s \geq 2$, the exponent 2^s is divisible by 4, so

$$5 \mid M.$$

Also M is odd and $M > 5$.

We have

$$n + 6 = 2^b + 4 = 4(2^{b-2} + 1) = 4(2^M + 1).$$

The divisor $5 \mid M$ implies

$$2^5 + 1 = 33 \mid 2^M + 1.$$

In particular,

$$11 \mid 2^M + 1,$$

and $11 > 6$.

On the other hand, Bang–Zsigmondy gives a primitive prime divisor R of

$$2^{2M} - 1.$$

Hence

$$\mathrm{ord}_R(2) = 2M,$$

so $R > 2M > 6$, and $R \mid 2^M + 1$.

Finally $R \neq 11$, since their multiplicative orders are different:

$$2M \neq 10.$$

Thus $n+6$ has two distinct prime-power components exceeding 6, namely 11 and R. Hence

$$V(n,6) \geq 2.$$

□

We have therefore obtained the following complete statement.

Theorem 5.4. *If*

$$V(n,k) \leq 1 \qquad \textit{for all } 1 \leq k \leq 6$$

and $d_2 = 1$, then

$$\boxed{n \in \{1,2,3,6,7,15,30\}.}$$

In particular, every counterexample with $n > 30$ satisfies

$$\boxed{d_2 = 2.}$$

Remark 5.5. *This theorem is stronger than the informal "Catalan + finite check" argument in earlier drafts. The infinite families which appear naturally after applying Catalan are eliminated using the order-theoretic form of the Bang–Zsigmondy theorem.*

6 Consequences of $d_2 = 2$

Assume now

$$d_2 = 2.$$

Then

$$n + 2 = 2q_2$$

with q_2 odd. Hence

$$n = 2(q_2 - 1),$$

and therefore

$$\boxed{4 \mid n.} \tag{11}$$

Since $d_3 \mid 6$ and $\gcd(d_2, d_3) = 1$,

$$\boxed{d_3 \in \{1,3\}.} \tag{12}$$

We now treat these two branches separately.

7 The branch $d_2 = 2,\ d_3 = 1$

Write

$$n + 3 = r^c.$$

Because $4 \mid n$,

$$r^c \equiv 3 \pmod 4,$$

so

$$\boxed{r \equiv 3 \pmod 4, \qquad c \text{ odd.}} \tag{13}$$

7.1 The $k = 4$ reduction

Set

$$s = v_p(r + 1).$$

Since c is odd, the 2-adic form of LTE gives

$$v_p(r^c + 1) = v_p(r + 1) = s. \tag{14}$$

7.1.1 Case $s = 2$

Then

$$r^c + 1 = 4A, \qquad A \text{ odd}.$$

Because $V(n, 4) \leq 1$, the odd part A can have at most one prime divisor exceeding 4. Since A itself is odd and has no factors 2, the large-prime condition forces A to be a power of one odd prime whenever $A > 1$.

Factor

$$A = \frac{r + 1}{4} \cdot \frac{r^c + 1}{r + 1} = A_1 B,$$

where

$$A_1 = \frac{r + 1}{4}, \qquad B = r^{c-1} - r^{c-2} + \cdots - r + 1.$$

If $r > 3$ and $c > 1$, then $A_1 > 1$ and $B > 1$. Since $A = A_1 B$ is a power of one prime ℓ,

$$A_1 = \ell^u, \qquad B = \ell^v.$$

In particular $\ell \mid r + 1$ and $\ell \mid B$. Since $r \equiv -1 \pmod{r + 1}$,

$$B \equiv c \pmod{r + 1},$$

hence

$$\ell \mid c.$$

By LTE,

$$v_\ell(B) = v_\ell(c).$$

But $B = \ell^{v_\ell(c)}$, and therefore

$$B \leq c.$$

On the other hand, for $r \geq 3$ and $c \geq 3$,

$$B = r^{c-1} - r^{c-2} + \cdots - r + 1 > c.$$

This contradiction shows:

$$\boxed{s = 2,\ c > 1 \Longrightarrow r = 3.} \tag{15}$$

7.1.2 Case $s \geq 3$

Now $2^s > 4$ itself is a large prime-power component of $n + 4$. Therefore the remaining odd part can involve no prime > 4. Since it is odd, the only possibility is 1 or 3:

$$r^c + 1 = 2^s \qquad \text{or} \qquad r^c + 1 = 3 \cdot 2^s.$$

If

$$r^c + 1 = 2^s$$

with $c > 1$, then

$$r^c + 1 = (r+1)(r^{c-1} - r^{c-2} + \cdots - r + 1)$$

has an odd factor > 1, impossible.

If

$$r^c + 1 = 3 \cdot 2^s,$$

then

$$(r+1)(r^{c-1} - r^{c-2} + \cdots - r + 1) = 3 \cdot 2^s.$$

The second factor is odd and > 1, so it must equal 3. But for $r \geq 3$ and $c \geq 3$,

$$r^{c-1} - r^{c-2} + \cdots - r + 1 \geq r^2 - r + 1 \geq 7,$$

again impossible.

Therefore

$$\boxed{s \geq 3,\ c > 1 \Longrightarrow \text{contradiction.}} \tag{16}$$

Combining (15) and (16) gives the key reduction.

Lemma 7.1. *If*

$$d_2 = 2, \qquad d_3 = 1, \qquad c > 1,$$

then

$$\boxed{r = 3}.$$

Consequently

$$\boxed{n = 3^c - 3.}$$

7.2 The exponent c is prime

Now

$$n + 2 = 3^c - 1 = 2q_2,$$

so

$$Q_c := \frac{3^c - 1}{2}$$

is a prime power.

Lemma 7.2. *If $c > 1$, then c is prime.*

Proof. Since c is odd, suppose

$$c = ab, \qquad a, b > 1.$$

Then

$$Q_c = \frac{3^a - 1}{2}(1 + 3^a + \cdots + 3^{a(b-1)}) =: AB.$$

Both A and B exceed 1. Since AB is a prime power,

$$A = \ell^u, \qquad B = \ell^v$$

for one prime ℓ.

As $\ell \mid 3^a - 1$,

$$B \equiv b \pmod{3^a - 1},$$

so

$$\ell \mid b.$$

By LTE,

$$v_\ell(B) = v_\ell(b).$$

Thus

$$B = \ell^{v_\ell(b)} \le b.$$

But

$$B = 1 + 3^a + \cdots + 3^{a(b-1)} > b,$$

contradiction. □

Hence

$$\boxed{c = p}$$

for some odd prime p, and

$$\boxed{n = 3^p - 3.} \tag{17}$$

At this point,

$$n + 1 = 3^p - 2 = P^\alpha, \tag{18}$$

$$n + 2 = 3^p - 1 = 2Q^\beta, \tag{19}$$

and

$$n + 4 = 3^p + 1 = 4R^\gamma. \tag{20}$$

In particular

$$\frac{3^p - 1}{2} = \Phi_p(3), \qquad \frac{3^p + 1}{4} = \Phi_{2p}(3),$$

so the $k = 2$ and $k = 4$ conditions impose prime-power values on two cyclotomic evaluations.

7.3 The decisive split modulo 4

7.3.1 Case $p \equiv 1 \pmod 4$

Consider

$$n + 6 = 3^p + 3 = 3(3^{p-1} + 1).$$

Because $p - 1$ is even,

$$v_p(3^{p-1} + 1) = 1.$$

Also

$$3^{p-1} + 1 \equiv 2 \pmod 3$$

and, since $p - 1 \equiv 0 \pmod 4$,

$$3^{p-1} + 1 \equiv 2 \pmod 5.$$

Therefore

$$T := \frac{3^{p-1} + 1}{2}$$

is odd and has no prime factors ≤ 5.

Since $V(n, 6) \le 1$,

$$\boxed{T \text{ is a prime power.}} \tag{21}$$

Write

$$p - 1 = 2^t m, \qquad m \text{ odd.}$$

Then

$$T = \frac{3^{2^t m} + 1}{2}.$$

The factor

$$A = \frac{3^{2^t} + 1}{2}$$

divides T.

Let ℓ be a prime divisor of A. Then

$$\mathrm{ord}_\ell(3) \mid 2^{t+1}.$$

By Bang–Zsigmondy, applied to

$$3^{2(p-1)} - 1,$$

there exists a primitive prime divisor R with

$$\mathrm{ord}_R(3) = 2(p-1) = 2^{t+1}m.$$

It divides

$$3^{p-1} + 1,$$

hence T. If $m > 1$, then

$$2^{t+1}m > 2^{t+1},$$

so $R \neq \ell$. Thus T has at least two distinct prime divisors, contradicting (21). Hence

$$m = 1,$$

so

$$p - 1 = 2^t. \tag{22}$$

Since $p > 3$ is prime, t is even. Thus

$$p = 2^t + 1 \equiv 2 \pmod 3.$$

Modulo 13, since $3^3 \equiv 1$,

$$3^p + 4 \equiv 3^2 + 4 \equiv 0 \pmod{13}.$$

Hence

$$13 \mid n + 7.$$

Moreover $n + 7$ is not divisible by $2, 3, 5, 7$:

$$n + 7 = 3^p + 4,$$

$$3^p + 4 \equiv 1 \pmod 3,$$

$$3^p + 4 \equiv 2 \pmod 5,$$

and $p \equiv 5 \pmod 6$ gives

$$3^p + 4 \equiv 3^5 + 4 \equiv 2 \pmod 7.$$

Therefore, under $V(n, 7) \leq 1$, the component 13 is the only possible large component, so

$$3^p + 4 = 13^\delta.$$

But modulo 8,

$$3^p + 4 \equiv 3 + 4 \equiv 7 \pmod 8,$$

whereas

$$13^\delta \equiv 1 \text{ or } 5 \pmod 8.$$

Contradiction.

Thus

$$\boxed{p \equiv 1 \pmod 4 \quad \Longrightarrow \quad \text{no counterexample.}} \tag{23}$$

7.3.2 Case $p \equiv 3 \pmod 4$

Since $p \equiv 3 \pmod 4$,

$$3^p - 2 \equiv 2 - 2 \equiv 0 \pmod 5.$$

But $n + 1 = 3^p - 2 = P^\alpha$ is a prime power. Therefore

$$P = 5,$$

so

$$3^p - 2 = 5^\alpha. \tag{24}$$

Modulo 8,

$$3^p - 2 \equiv 3 - 2 \equiv 1 \pmod 8.$$

Since

$$5^\alpha \equiv 1 \pmod 8$$

only when α is even,

$$\alpha = 2a.$$

Thus

$$(5^a)^2 + 2 = 3^p.$$

We now invoke Ljunggren's classical theorem on the generalized Lebesgue–Nagell equation: for integers $x > 0$, $y > 1$, $m > 1$,

$$x^2 + 2 = y^m$$

has only the solution

$$(x, y, m) = (5, 3, 3).$$

See [6].

Therefore

$$a = 1, \qquad p = 3.$$

Consequently

$$\boxed{n = 3^3 - 3 = 24.}$$

We have thus proved:

Proposition 7.3. *If*

$$d_2 = 2, \qquad d_3 = 1,$$

and the exponent in

$$n + 3 = r^c$$

satisfies $c > 1$, then

$$\boxed{n = 24.}$$

This closes the entire $d_2 = 2, d_3 = 1, c > 1$ branch.

8 Analysis and reduction of the branch $d_2 = 2,\ d_3 = 1$

We now strengthen the analysis of the leaf $d_2 = 2, d_3 = 1$. The proper prime-power case $n + 3 = r^c$, $c > 1$, was settled in Proposition 7.3. We complete the complementary case $c = 1$ here.

Assume throughout this section that

$$V_1(n) = 1, \qquad d_2 = 2, \qquad d_3 = 1, \qquad n + 3 = R$$

with R prime. Then

$$n + 1 = P^\alpha, \qquad n + 2 = 2Q^\beta,$$

where P, Q are odd primes. Equivalently,

$$\boxed{P^\alpha + 1 = 2Q^\beta, \qquad P^\alpha + 2 = R.} \tag{25}$$

The condition $d_2 = 2$ implies $4 \mid n$.

Proposition 8.1 (Exponent rigidity on the prime $n + 3$ leaf)**.** *Assume*

$$V_1(n) = 1, \qquad d_2 = 2, \qquad d_3 = 1, \qquad n + 3 = R$$

with R prime. Write

$$n + 1 = P^\alpha, \qquad n + 2 = 2Q^\beta.$$

Then the following reductions hold.

(i) If $\alpha > 1$, then

$$P = 3, \qquad \beta = 1, \qquad \alpha \in \{2, 4\},$$

and hence

$$n \in \{8, 80\}.$$

(ii) If $\alpha = 1$ and $\beta > 1$, then

$$Q = 3, \qquad \beta = 2^s \quad (s \geq 1),$$

and necessarily

$$2 \cdot 3^\beta - 1, \qquad \frac{3^\beta + 1}{2}, \qquad 2 \cdot 3^\beta + 1$$

are all prime. The case $s = 1$ gives $n = 16$. The elimination of $s > 1$ is left as an explicit residual Diophantine problem.

(iii) If $\alpha = \beta = 1$, then $n = 4$.

Consequently, apart from the explicit residual family in (ii), the $d_2 = 2, d_3 = 1$ branch consists only of

$$\boxed{n \in \{4, 8, 16, 24, 80\}}.$$

Proof. Suppose first that an odd prime s divides α, and write $\alpha = st$ with $s > 1$ odd. Then

$$P^\alpha + 1 = (P^t + 1)\left(P^{t(s-1)} - P^{t(s-2)} + \cdots - P^t + 1\right).$$

The second factor is odd and greater than 1. Since the product is $2Q^\beta$, the odd part of the first factor and the second factor can involve only the prime Q. Put $X = P^t$ and

$$B = \frac{X^s + 1}{X + 1}.$$

Then $Q \mid X+1$ and, reducing modulo $X+1$,

$$B \equiv s \pmod{X+1}.$$

Hence $Q \mid s$. Since s is odd, LTE gives

$$\nu_Q(B) = \nu_Q(s).$$

Thus $B \le s$, whereas $B > s$ for $X \ge 3$ and $s \ge 3$, a contradiction. Therefore

$$\boxed{\alpha = 2^u}.$$

Assume now $\alpha > 1$ and put

$$X = P^{\alpha/2}.$$

Then

$$X^2 + 1 = 2Q^\beta, \qquad R = X^2 + 2.$$

Since R is prime, $P \neq 3$ would give $X^2 \equiv 1 \pmod 3$ and hence $R \equiv 0 \pmod 3$, impossible for $R > 3$. Thus

$$P = 3.$$

If $\beta > 2$, the classification of

$$x^2 + 1 = 2y^m$$

for $xy > 1$ and $m > 2$ gives only $(x, y, m) = (239, 13, 4)$; see [4, 5]. This is impossible because X is a power of 3. If $\beta = 2$, then $3 \mid X$ and reduction modulo 3 gives

$$1 \equiv 2Q^2 \pmod 3,$$

which is impossible. Hence

$$\boxed{\beta = 1}.$$

Consequently

$$n = 3^\alpha - 1, \qquad \alpha = 2^u.$$

Suppose $\alpha > 2$. Since

$$n + 4 = 3(3^{\alpha-1} + 1),$$

the condition $V(n, 4) \le 1$ implies that

$$T = \frac{3^{\alpha-1} + 1}{4}$$

is a prime power. Put $m = \alpha - 1 = 2^u - 1$. If $m = rs$ with $r, s > 1$ odd, then, with $x = 3^r$,

$$T = \frac{x+1}{4}\left(x^{s-1} - x^{s-2} + \cdots - x + 1\right).$$

The two factors are > 1. If their product is a power of one odd prime ℓ, then $\ell \mid x + 1$ and the second factor is congruent to s modulo $x + 1$. Hence $\ell \mid s$, and LTE gives its ℓ-adic valuation as $\nu_\ell(s)$. It is therefore at most s, whereas the factor itself is $> s$. Contradiction. Hence m is prime, and so u is prime.

If u is odd and $u \ge 3$, then $\alpha = 2^u$ satisfies $\alpha \equiv 0 \pmod 4$ and $\alpha \equiv 2 \pmod 3$. Consequently

$$5 \mid 3^\alpha + 4, \qquad 13 \mid 3^\alpha + 4.$$

Let

$$M = 3^\alpha + 4.$$

Since $M \equiv 7 \pmod 8$, M cannot have the form $5^a 13^b$, because $5^a 13^b \equiv 1$ or $5 \pmod 8$. Thus, besides the divisor 13, there is another prime divisor exceeding 5. The two corresponding prime-power components both exceed the offset $k = 5$, so

$$V(n, 5) \geq 2,$$

a contradiction. Therefore $u = 1$ or $u = 2$, giving

$$\alpha = 2, 4, \qquad n = 8, 80.$$

This proves (i).

Now assume $\alpha = 1$ and $\beta > 1$. Since P and $P + 2$ are both prime, $P \not\equiv 1 \pmod 3$, so $P \equiv 2 \pmod 3$. Hence $3 \mid P + 1 = 2Q^\beta$, forcing

$$Q = 3.$$

Thus

$$P = 2 \cdot 3^\beta - 1, \qquad P + 2 = 2 \cdot 3^\beta + 1.$$

If β is odd, then according as $\beta \equiv 1$ or $3 \pmod 4$, one of these two primes is divisible by 5. Hence $\beta = 2r$ is even.

At $k = 4$,

$$n + 4 = 4\frac{3^\beta + 1}{2},$$

so

$$T = \frac{3^\beta + 1}{2}$$

is a prime power. Put $X = 3^r$. Then

$$X^2 + 1 = 2Y^e.$$

The same classification of $x^2 + 1 = 2y^m$ rules out $e > 2$ except $(239, 13, 4)$, impossible for $X = 3^r$; and $e = 2$ is impossible modulo 3. Hence $e = 1$, so T is prime.

If r has an odd divisor $s > 1$, then $3^{2r} + 1$ factors as

$$(3^{2r/s} + 1)\left(3^{2r(s-1)/s} - 3^{2r(s-2)/s} + \cdots - 3^{2r/s} + 1\right),$$

so T is composite. Therefore $r = 2^w$, and hence

$$\boxed{\beta = 2^{w+1}}.$$

The case $w = 0$ gives $\beta = 2$, hence $P = 17$ and $n = 16$. For $w > 0$ the three quantities

$$2 \cdot 3^\beta - 1, \qquad \frac{3^\beta + 1}{2}, \qquad 2 \cdot 3^\beta + 1$$

must all be prime. We leave this simultaneous-primality family open. This proves (ii).

Finally suppose $\alpha = \beta = 1$. Put

$$q = \frac{n + 2}{2}.$$

Then

$$n + 1 = 2q - 1, \qquad n + 3 = 2q + 1$$

are both prime. If $q > 3$, one of these two numbers is divisible by 3 and is strictly larger than 3. Hence $q = 3$, giving $n = 4$. □

Corollary 8.2 (Complete $d_2 = 2, d_3 = 1$ branch)**.** *If* $V_1(n) = 1$ *and*

$$d_2 = 2, \qquad d_3 = 1,$$

then

$$\boxed{n \in \{4, 8, 16, 24, 80\}}.$$

Indeed, $c > 1$ *gives* $n = 24$ *by Proposition 7.3, while* $c = 1$ *gives* $n \in \{4, 8, 16, 80\}$ *by Proposition 8.1.*

Remark 8.3 (Interpretation)**.** *The corollary classifies an entire state of the* d_k *decision tree. Thus any counterexample with* $n > 80$ *must lie in the branch*

$$\boxed{d_2 = 2, \qquad d_3 = 3.}$$

This concentrates the remaining Diophantine work on the $d_3 = 3$ *branch.*

8.1 The residual simultaneous-primality family

The unresolved part of Proposition 8.1 has a particularly compact one-parameter form. For $s > 1$ put

$$\beta = 2^s.$$

Then any remaining counterexample on the prime $n + 3$ leaf must satisfy

$$\boxed{2 \cdot 3^{2^s} - 1, \qquad \frac{3^{2^s} + 1}{2}, \qquad 2 \cdot 3^{2^s} + 1 \text{ are prime.}} \tag{41}$$

The first and third expressions are the two endpoints $n+1$ and $n+3$, while the middle expression is forced by the $k = 4$ condition. This turns the unresolved leaf into a simultaneous-primality problem in a single power-of-two exponent. The OEIS records the exponents for which $2 \cdot 3^m - 1$ is prime and separately those for which $2 \cdot 3^m + 1$ is prime; these sequences show that the remaining question is not a routine congruence exercise. We therefore treat (41) as a genuine residual Diophantine target rather than assert an unsupported elimination.

Proposition 8.4 (Local obstructions for the residual prime triple)**.** *Let* $s > 1$*. If all three numbers in* (41) *are prime, then*

$$\boxed{s \equiv 5 \text{ or } 9 \pmod{12}, \qquad s \not\equiv 9 \pmod{11}, \qquad s \not\equiv 9 \pmod{10}, \qquad s \not\equiv 5 \pmod{20}.}$$

Consequently, the residual family has no solution with $2 \le s \le 16$*; the least parameter not excluded by these congruences is* $s = 17$*.*

Proof. Put $x = 3^{2^s}$, and write

$$A = 2x - 1, \qquad C = 2x + 1.$$

Both A and C are prime by hypothesis.

If s is even, then $2^s \equiv 4 \pmod 6$, so

$$x \equiv 3^4 \equiv 4 \pmod 7, \qquad A \equiv 0 \pmod 7.$$

Since $A \ge 161 > 7$, this contradicts primality. Thus s is odd.

If $s \equiv 3 \pmod 4$, then $2^s \equiv 3 \pmod 5$. As $3^5 \equiv 1 \pmod{11}$,

$$x \equiv 3^3 \equiv 5 \pmod{11}, \qquad C \equiv 0 \pmod{11}.$$

Here $C > 11$, a contradiction. Hence $s \not\equiv 3 \pmod 4$.

If $s \equiv 1 \pmod 6$, then $2^s \equiv 2 \pmod{18}$: indeed, $2^s \equiv 2 \pmod 9$ and 2^s is even. Fermat's theorem gives

$$x \equiv 3^2 \equiv 9 \pmod{19}, \qquad C \equiv 0 \pmod{19}.$$

Because $s > 1$, such an s is at least 7, so $C > 19$, again a contradiction. Combining the three restrictions obtained so far yields

$$s \equiv 5 \text{ or } 9 \pmod{12}.$$

For the remaining exclusions, the same argument applies to A. If $s \equiv 9 \pmod{11}$, then $2^s \equiv 6 \pmod{23}$ and $3^{2^s} \equiv 3^6 \equiv 24 \pmod{47}$, so $47 \mid A$. Since $A > 47$, this is impossible.

If $s \equiv 9 \pmod{10}$, then $s \geq 9$, and the exponent 2^s is congruent to 0 (mod 8) and 6 (mod 11), hence to 72 (mod 88). Thus

$$x \equiv 3^{72} \equiv 45 \pmod{89}, \qquad A \equiv 0 \pmod{89}.$$

Again $A > 89$, a contradiction.

Finally, if $s \equiv 5 \pmod{20}$, then $2^s \equiv 32 \pmod{155}$. Since $3^{155} \equiv 1 \pmod{311}$ and $3^{32} \equiv 156 \pmod{311}$,

$$x \equiv 156 \pmod{311}, \qquad A \equiv 0 \pmod{311}.$$

Such an $s > 1$ is at least 5, so $A > 311$, which contradicts the primality of A. This proves all the claimed restrictions.

For $2 \leq s \leq 16$, an even s is ruled out by 7; among the odd values, $s \equiv 3 \pmod 4$, $s \equiv 1 \pmod 6$, or one of the last three forbidden classes applies. Directly, the only values in this range left after the first three restrictions are 5 and 9, which are excluded by 311 and 47, respectively. The congruence conditions all hold at $s = 17$, so 17 is the least parameter not ruled out by this finite local sieve. □

Since $n = 2 \cdot 3^{2^s} - 2$ on this leaf, Proposition 8.4 shows that any solution in this residual family would satisfy

$$n \geq 2 \cdot 3^{131072} - 2.$$

This is only a lower bound for a possible solution; it does not eliminate the remaining simultaneous-primality family.

9 The branch $d_2 = 2,\ d_3 = 3$

We now assume

$$d_3 = 3.$$

Then

$$n + 3 = 3q_3,$$

so $3 \mid n$. Together with $4 \mid n$,

$$\boxed{12 \mid n.} \tag{26}$$

Since

$$d_4 \mid \operatorname{lcm}(1, 2, 3, 4) = 12$$

and $\gcd(d_3, d_4) \mid 1$, we have

$$3 \nmid d_4.$$

Thus

$$d_4 \in \{1, 2, 4\}.$$

But $4 \mid n$ implies $4 \mid n + 4$, so the complete 2-adic small component cannot be represented by $d_4 = 2$. Hence

$$\boxed{d_4 \in \{1, 4\}.} \tag{27}$$

9.1 The subbranch $d_4 = 1$

If

$$d_4 = 1,$$

then

$$n + 4 = 2^a.$$

Since $3 \mid n$,

$$2^a \equiv 1 \pmod 3,$$

so a is even:

$$a = 2b.$$

Then

$$n + 3 = 2^{2b} - 1,$$

and because $d_3 = 3$,

$$\frac{2^{2b} - 1}{3}$$

must be a prime power.

But

$$\frac{2^{2b} - 1}{3} = \begin{cases} (2^b - 1)\dfrac{2^b + 1}{3}, & b \text{ odd}, \\ \dfrac{2^b - 1}{3}(2^b + 1), & b \text{ even}. \end{cases} \tag{28}$$

For $b > 2$, the two factors are integers > 1 and are coprime. Hence the product cannot be a prime power.

Therefore

$$\boxed{d_4 = 1 \Longrightarrow n = 12.} \tag{29}$$

Consequently, for every $n > 12$ in this branch,

$$\boxed{d_4 = 4.} \tag{30}$$

Lemma 9.1 (The $d_4 = 4$ branch is divisible by 24)**.** *Assume $d_2 = 2$, $d_3 = 3$, and $d_4 = 4$. Then*

$$\boxed{24 \mid n}.$$

Consequently, if $d_5 = 5$, then $120 \mid n$.

Proof. We know $12 \mid n$, so write $n = 12m$. Since

$$q_4 = \frac{n + 4}{4} = 3m + 1$$

is coprime to $d_4 = 4$, the integer q_4 is odd. Hence m is even, so $24 \mid n$. If $d_5 = 5$, then also $5 \mid n$, giving $120 \mid n$. □

9.2 The two subbranches for d_5

Since $12 \mid n$,

$$n + 5$$

is odd and not divisible by 3. Moreover

$$d_5 \mid \operatorname{lcm}(1, \ldots, 5) = 60$$

and

$$\gcd(d_4, d_5) = 1.$$

Hence

$$\boxed{d_5 \in \{1, 5\}.} \tag{31}$$

9.2.1 Subbranch $d_5 = 1$

Here

$$n + 5 = p$$

is prime. Thus

$$\begin{aligned} n + 1 &= p - 4 = P^{\alpha}, \\ n + 2 &= p - 3 = 2Q^{\beta}, \\ n + 3 &= p - 2 = 3R^{\gamma}, \\ n + 4 &= p - 1 = 4S^{\delta}. \end{aligned}$$

Hence

$$\boxed{\begin{aligned} p - 4 &= P^{\alpha}, \\ p - 3 &= 2Q^{\beta}, \\ p - 2 &= 3R^{\gamma}, \\ p - 1 &= 4S^{\delta}, \end{aligned}} \tag{32}$$

with p prime.

Since $24 \mid n = p - 5$ by Lemma 9.1,

$$\boxed{p \equiv 5 \pmod{24}.} \tag{33}$$

The only solution currently visible in the finite computation with the required d-pattern is

$$p = 53, \qquad n = 48.$$

Indeed,

$$\begin{aligned} 53 - 4 &= 49 = 7^2, \\ 53 - 3 &= 50 = 2 \cdot 5^2, \\ 53 - 2 &= 51 = 3 \cdot 17, \\ 53 - 1 &= 52 = 4 \cdot 13. \end{aligned}$$

The superficially similar value $p = 29$ corresponds to $n = 24$, but there $n + 3 = 27 = 3^3$, so it belongs to the already classified $d_3 = 1$ branch.

At present we do not have a proof that these are the only solutions to (32). This is one of the principal remaining Diophantine targets.

9.2.2 Subbranch $d_5 = 5$

Here

$$\boxed{120 \mid n}$$

by Lemma 9.1. Write

$$n = 120u.$$

Then

$$\begin{aligned} n + 1 &= 120u + 1 = q_1, \\ n + 2 &= 2(60u + 1) = 2q_2, \\ n + 3 &= 3(40u + 1) = 3q_3, \\ n + 4 &= 4(30u + 1) = 4q_4, \\ n + 5 &= 5(24u + 1) = 5q_5. \end{aligned} \tag{34a}$$

Each q_i is a prime power, and the small factors displayed explicitly are coprime to their corresponding q_i.

Subtracting consecutive equations gives the exact prime-power chain

$$\boxed{\begin{aligned} 2q_2 - q_1 &= 1, \\ 3q_3 - 2q_2 &= 1, \\ 4q_4 - 3q_3 &= 1, \\ 5q_5 - 4q_4 &= 1. \end{aligned}} \tag{34}$$

This is one of the main Diophantine objects remaining in the project.

9.3 The $k = 6$ split

Because

$$n + 6 = 120u + 6 = 6(20u + 1),$$

and $d_6 \mid 120$ with $\gcd(d_5, d_6) = 1$, the only possibilities for a large counterexample are

$$\boxed{d_6 \in \{2, 6\}.} \tag{35}$$

Indeed, $d_6 = 1$ leaves the distinct factors 2 and 3 in the remainder, while $d_6 = 3$ leaves an even nontrivial remainder and hence cannot produce a single prime power; the cases containing 5 are excluded by $\gcd(d_5, d_6) = 1$.

9.3.1 Case $d_6 = 6$

Then

$$q_6 = 20u + 1$$

is a prime power and

$$\boxed{6q_6 - 5q_5 = 1.} \tag{36}$$

Together with (34), this yields the six-term chain

$$\boxed{120u + 1,\ 60u + 1,\ 40u + 1,\ 30u + 1,\ 24u + 1,\ 20u + 1} \tag{37}$$

of prime-power expressions. This branch remains unresolved.

9.3.2 Case $d_6 = 2$

Now

$$n + 6 = 6(20u + 1).$$

Since $d_6 = 2$, the remaining component is

$$q_6 = 3(20u + 1).$$

For this to be a prime power, there must be an integer $b \geq 1$ such that

$$20u + 1 = 3^b.$$

The congruence modulo 20 gives

$$3^b \equiv 1 \pmod{20},$$

so

$$\boxed{b \equiv 0 \pmod{4}.}$$

Write

$$b = 4t, \qquad u = \frac{3^{4t} - 1}{20}. \tag{38}$$

Set

$$N = 4t + 1.$$

Then the five prime-power quantities forced by $k = 1, \ldots, 5$ are

$$\boxed{\begin{aligned} q_1 &= 2 \cdot 3^N - 5, \\ q_2 &= 3^N - 2, \\ q_3 &= 2 \cdot 3^{N-1} - 1, \\ q_4 &= \frac{3^N - 1}{2}, \\ q_5 &= \frac{2 \cdot 3^N - 1}{5}. \end{aligned}} \tag{40}$$

Each of these five quantities must be a prime power.

Lemma 9.2 (Prime-index reduction)**.** *In the branch $d_5 = 5, d_6 = 2$, the integer $N = 4t + 1$ is prime.*

Proof. Suppose $N = rs$ with $r, s > 1$ odd, and put $x = 3^r$. Then

$$\frac{3^N - 1}{2} = \frac{x - 1}{2}(1 + x + \cdots + x^{s-1}).$$

Both factors exceed 1. If their product is a power of one prime ℓ, then ℓ divides both factors. Hence $\ell \mid x - 1$, while the second factor is congruent to s modulo $x - 1$, so $\ell \mid s$. By LTE,

$$\nu_\ell(1 + x + \cdots + x^{s-1}) = \nu_\ell(s).$$

Thus the second factor is at most s, contradicting

$$1 + x + \cdots + x^{s-1} > s.$$

Hence N is prime. □

Consequently this leaf reduces to the simultaneous prime-power system (40) with

$$\boxed{N \equiv 1 \pmod 4, \qquad N \text{ prime.}}$$

We can sharpen the congruence class of N further.

Lemma 9.3 (A mod-12 restriction)**.** *In the branch $d_5 = 5, d_6 = 2$, one has*

$$\boxed{N \equiv 1 \pmod{12}.}$$

Proof. Since N is prime and $N > 3$, we have

$$N \equiv 1 \text{ or } 5 \pmod 6.$$

Suppose $N \equiv 5 \pmod 6$. Then

$$N - 1 \equiv 4 \pmod 6,$$

and hence, modulo 7,

$$3^{N-1} \equiv 3^4 \equiv 4 \pmod 7.$$

Therefore

$$q_3 = 2 \cdot 3^{N-1} - 1 \equiv 7 \equiv 0 \pmod 7.$$

But $q_3 > 7$ for $N > 3$, while

$$q_3 \equiv -1 \equiv 2 \pmod 3.$$

Thus q_3 cannot be a power of 7, since every positive power of 7 is 1 modulo 3. This contradicts the prime-power condition on q_3. Consequently

$$N \equiv 1 \pmod 6.$$

Together with $N \equiv 1 \pmod 4$, this gives

$$N \equiv 1 \pmod{12}.$$

□

Thus the current sharp form of the $d_5 = 5, d_6 = 2$ branch is

$$\boxed{N \text{ prime}, \qquad N \equiv 1 \pmod{12},}$$

together with the five simultaneous prime-power conditions (40). This is the sharpest form of the branch obtained so far.

9.4 Targeted computation of the residual $d_6 = 2$ leaf

The residual family (40) is sufficiently structured that small exact computations are useful for guiding the next proof. The table below records the first few admissible prime indices $N = 4t+1$ and a first prime-power condition that fails.

t	N	first obstruction
1	5	$q_1 = 481 = 13 \cdot 37$
3	13	$q_2 = 1594321 = 197 \cdot 8093$
4	17	$q_1 = 2 \cdot 3^{17} - 5 = 13 \cdot 71 \cdot 461 \cdot 607$
7	29	$q_1 = 13^2 \cdot 139 \cdot 5843120971$

This table is computational evidence only. It does, however, illustrate that the five simultaneous prime-power requirements in (40) are significantly stronger than any one of them individually.

10 The current decision tree

The proved reductions can be summarized as follows.

$$\begin{array}{ccc} & V_1(n) = 1 & \\ & \downarrow & \\ d_2 = 1 & \text{or} & d_2 = 2 \end{array}$$

The first branch is completely closed:

$$\boxed{d_2 = 1 \Longrightarrow n \in \{1, 2, 3, 6, 7, 15, 30\}.}$$

For $n > 30$,

$$d_2 = 2.$$

Then

$$d_3 \in \{1, 3\}.$$

For $d_3 = 1$, write

$$n + 3 = r^c.$$

The case $c > 1$ is completely closed:

$$\boxed{d_2 = 2,\ d_3 = 1,\ c > 1 \Longrightarrow n = 24.}$$

Therefore every large unresolved point on this branch must have

$$\boxed{n + 3 \text{ prime.}}$$

For $d_3 = 3$,

$$12 \mid n.$$

Then

$$d_4 \in \{1, 4\},$$

with

$$d_4 = 1 \Longrightarrow n = 12.$$

Hence for $n > 12$,

$$d_4 = 4,$$

and then

$$d_5 \in \{1, 5\}.$$

The resulting research tree is therefore

$$\begin{array}{rcl}
d_2 = 1 & \Longrightarrow & n \in \{1, 2, 3, 6, 7, 15, 30\}, \\
d_2 = 2,\ d_3 = 1,\ c > 1 & \Longrightarrow & n = 24, \\
d_2 = 2,\ d_3 = 1,\ c = 1 & \Longrightarrow & n \in \{4, 8, 16, 80\} \text{ or the residual family (41)}, \\
d_2 = 2,\ d_3 = 3,\ d_4 = 1 & \Longrightarrow & n = 12, \\
d_2 = 2,\ d_3 = 3,\ d_4 = 4,\ d_5 = 1 & \Longrightarrow & \text{system (32)}, \\
d_2 = 2,\ d_3 = 3,\ d_4 = 4,\ d_5 = 5 & \Longrightarrow & \text{chain (34) and families (37)–(38)}.
\end{array} \tag{39}$$

This is the main structural output of the present work.

11 Cyclotomic interpretation

The branch

$$d_2 = 2, \qquad d_3 = 1, \qquad c > 1$$

provides a particularly clear cyclotomic interpretation.

After proving

$$n = 3^p - 3,$$

the conditions at $k = 2$ and $k = 4$ become

$$\frac{3^p - 1}{2} = \Phi_p(3)$$

and

$$\frac{3^p + 1}{4} = \Phi_{2p}(3).$$

Thus a counterexample would force both

$$\Phi_p(3) \quad \text{and} \quad \Phi_{2p}(3)$$

to be prime powers, while simultaneously

$$3^p - 2$$

would have to be a prime power.

The proof above shows how the requirement at $k = 6$ forces strong restrictions on the exponent p, and the requirement at $k = 7$ then creates the congruence obstruction

$$13 \mid 3^p + 4.$$

In the remaining residue class, the divisibility

$$5 \mid 3^p - 2$$

converts the problem into the classical equation

$$x^2 + 2 = 3^p.$$

This is a useful model for future work: large counterexamples may be forced into simultaneous primality or prime-power conditions on several nearby cyclotomic values, after which primitive-divisor theory becomes relevant.

12 Proof ledger: what is proved and what is not

For clarity, we record the logical status of the main statements.

Proved in this manuscript

1. The prime-power-component reformulation

$$V(n,k) \geq 2 \iff Q_2(n+k) > k.$$

2. The offset bound

$$V(n,k) \geq 2 \Longrightarrow k^2 - k < n.$$

3. The divisor decomposition

$$n + k = d_k q_k, \qquad d_k \mid L_k.$$

4. The gcd compatibility

$$\gcd(d_i, d_j) \mid |i - j|.$$

5. The complete elimination of $d_2 = 1$, apart from

$$1, 2, 3, 6, 7, 15, 30.$$

6. The implication

$$d_2 = 2 \Longrightarrow 4 \mid n.$$

7. The implication

$$d_2 = 2,\ d_3 = 1,\ c > 1 \Longrightarrow r = 3,\ c \text{ prime}.$$

8. The complete elimination of that latter branch down to

$$n = 24.$$

9. The implications
$$d_2 = 2,\ d_3 = 3 \Longrightarrow 12 \mid n,$$
$$d_4 \in \{1, 4\},$$
and
$$d_4 = 1 \Longrightarrow n = 12.$$

10. For $n > 12$ on the $d_3 = 3$ branch,
$$d_4 = 4, \qquad d_5 \in \{1, 5\}.$$

11. The prime-power reduction for the $d_5 = 1$ and $d_5 = 5$ leaves.

12. The exponential family arising from $d_6 = 2$.

Computationally established only

The exact finite computation up to
$$5 \cdot 10^7$$
and the distribution in Table 1 are computational statements. They provide evidence and useful targets, but they do not prove finiteness.

Still open within the present project

The principal unresolved systems are:

1. The four-consecutive-prime-power system
$$p - 4 = P^{\alpha}, \quad p - 3 = 2Q^{\beta}, \quad p - 2 = 3R^{\gamma}, \quad p - 1 = 4S^{\delta}.$$

2. The prime-power chain
$$120u + 1,\ 60u + 1,\ 40u + 1,\ 30u + 1,\ 24u + 1,\ 20u + 1$$
together with the relations (34) and (36).

3. The exponential family (38).

These are the natural next targets for further work.

13 Research directions

The present reductions suggest several concrete directions.

13.1 Classifying the small factors d_k

The condition
$$\gcd(d_i, d_j) \mid |i - j|$$
can be interpreted as a compatibility constraint on the prime support of the sequence
$$d_1, d_2, d_3, \ldots.$$
For small k, each d_k is a divisor of a known least common multiple. The goal is to classify all compatible patterns before introducing the large prime-power components q_k.

This produces a finite-state type of problem: each prime ℓ can enter the small part only for a controlled range of offsets, and repeated use of the same prime is forced to respect the differences $i - j$.

13.2 Primitive divisors of the large components

The successful elimination of several infinite families used the following principle.

Suppose

$$A \mid B$$

are exponents and both

$$a^A + 1, \qquad a^B + 1$$

occur inside the same prime-power constraint. A primitive divisor at level $2B$ has multiplicative order $2B$, and therefore cannot coincide with a primitive divisor at a proper divisor level.

This makes Bang–Zsigmondy particularly well suited to the present problem: prime-power hypotheses are incompatible with the existence of primitive divisors at two different levels.

13.3 Prime-power chains

The relation

$$a_1 q_1 - a_2 q_2 = 1$$

between prime powers is extremely restrictive. The branch $d_5 = 5$ produces four such relations immediately, and $k = 6$ adds another. One possible approach is to combine congruences modulo carefully chosen primes with valuation arguments and primitive divisor results.

13.4 The consecutive-prime-power leaf

The system

$$p - 4 = P^\alpha, \quad p - 3 = 2Q^\beta, \quad p - 2 = 3R^\gamma, \quad p - 1 = 4S^\delta$$

remains one of the cleanest surviving Diophantine problems in the $d_3 = 3, d_4 = 4, d_5 = 1$ branch. The finite computation identifies the configuration $p = 53$ (equivalently $n = 48$) among the small cases. The value $p = 29$ corresponds instead to $n = 24$, for which $n + 3 = 27 = 3^3$ and therefore belongs to the already classified $d_3 = 1$ branch.

A useful next theorem would be a nontrivial restriction on the exponents

$$\alpha, \beta, \gamma, \delta,$$

for example a parity theorem, a forced exponent-1 statement, or a bound under natural coprimality assumptions.

14 Conclusion

The prime-power variant of Erdős–Selfridge Problem #889 remains open. The main contribution of the present draft is a rigorous structural reduction, not a complete resolution.

The divisor decomposition

$$n + k = d_k q_k$$

and the compatibility law

$$\gcd(d_i, d_j) \mid |i - j|$$

convert the original problem into a controlled collection of exponential Diophantine configurations. The entire $d_2 = 1$ branch can be closed using Catalan–Mihăilescu and Bang–Zsigmondy, while the substantial branch

$$d_2 = 2, \ d_3 = 1, \ c > 1$$

reduces completely to the exceptional value $n = 24$, with the crucial final step supplied by the classical equation

$$x^2 + 2 = 3^p.$$

The remaining problem is therefore considerably narrower than the original formulation. The main open tasks are now explicit systems of equations between prime powers, together with the compatible small-factor sequence (d_k). This is the point at which the investigation can move from global experimentation to the classification of individual Diophantine leaves.

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A Exact computational sieve

The principal computational object is the second-largest prime-power component $Q_2(m)$. The following implementation uses a linear smallest prime factor sieve and then accumulates the intervals (1).

```
#include <bits/stdc++.h>
using namespace std;

// Exact search for the Erdős-Selfridge prime-power variant:
//
// V(n,k) = number of p such that p^{nu_p(n+k)} > k.
```

```
//
// We seek all n <= N with max_{k>=1} V(n,k) <= 1.
//
// Let Q2(m) be the second-largest prime-power component of m.
// Then V(n,k) >= 2 iff Q2(n+k) > k.
// Hence m=n+k covers n precisely when
//
//     m - Q2(m) + 1 <= n <= m-1.
//
// Also, V(n,k)>=2 implies k^2-k<n.

static int Kbound(long long N) {
    long long k = 1;
    while ((k + 1) * (k + 1) - (k + 1) < N) ++k;
    return (int)k;
}

int main(int argc, char** argv) {
    int N = (argc >= 2 ? atoi(argv[1]) : 1000000);
    if (N < 1) return 1;

    const int K = Kbound(N);
    const int M = N + K;

    cerr << "N=" << N << " K=" << K << " M=" << M << '\n';

    // Smallest prime factor table via a linear sieve.
    vector<uint32_t> spf(M + 1, 0);
    vector<int> primes;
    primes.reserve(M / max(1, (int)log((double)M)));

    for (int i = 2; i <= M; ++i) {
        if (spf[i] == 0) {
            spf[i] = (uint32_t)i;
            primes.push_back(i);
        }

        for (int p : primes) {
            if (p > (int)spf[i] || 1LL * i * p > M) break;
            spf[i * p] = (uint32_t)p;
        }
    }

    // q2[m] = second-largest prime-power component of m.
    vector<uint32_t> q2(M + 1, 0);

    for (int m = 2; m <= M; ++m) {
        int x = m;
        uint64_t best1 = 0, best2 = 0;

        while (x > 1) {
```

```
        int p = (int)spf[x];
        uint64_t q = p;

        x /= p;
        while (x > 1 && (int)spf[x] == p) {
            x /= p;
            q *= (uint64_t)p;
        }

        if (q > best1) {
            best2 = best1;
            best1 = q;
        } else if (q > best2) {
            best2 = q;
        }
    }

    q2[m] = (uint32_t)best2;
}

// Difference-array interval coverage.
vector<int32_t> diff(N + 2, 0);

for (int m = 2; m <= M; ++m) {
    uint32_t q = q2[m];
    if (q == 0) continue;

    int L = max(1, m - (int)q + 1);
    int R = min(N, m - 1);

    if (L <= R) {
        ++diff[L];
        --diff[R + 1];
    }
}

vector<int> exceptions;
int coverage = 0;

for (int n = 1; n <= N; ++n) {
    coverage += diff[n];
    if (coverage == 0) exceptions.push_back(n);
}

cout << "exceptions=" << exceptions.size() << '\n';
cout << "last="
     << (exceptions.empty() ? -1 : exceptions.back()) << '\n';

for (size_t i = 0; i < exceptions.size(); ++i) {
    if (i) cout << ' ';
    cout << exceptions[i];
```

```
    }
    cout << '\n';

    vector<long long> hist(32, 0);
    int max_k = 0, max_n = -1;

    unordered_set<int> is_exception;
    is_exception.reserve(exceptions.size() * 2 + 1);

    for (int n : exceptions) is_exception.insert(n);

    for (int n = 1; n <= N; ++n) {
        if (is_exception.count(n)) continue;

        int witness = -1;

        for (int k = 1; k <= K; ++k) {
            if (q2[n + k] > (uint32_t)k) {
                witness = k;
                break;
            }
        }

        if (witness < 0) {
            cerr << "ERROR: no witness for n=" << n << '\n';
            return 2;
        }

        if (witness >= (int)hist.size())
            hist.resize(witness + 1);

        ++hist[witness];

        if (witness > max_k) {
            max_k = witness;
            max_n = n;
        }
    }

    cout << "max_min_k=" << max_k
         << " at_n=" << max_n << '\n';

    for (int k = 1; k <= max_k; ++k) {
        if (hist[k])
            cout << k << ':' << hist[k] << ' ';
    }

    cout << '\n';
}
```

B Selected arithmetic identities

For convenient reference, the main identities used in the reductions are collected here.

Branch $d_2 = 1$

$$n + 1 = 2^a, \qquad n + 2 = 2^a + 1,$$

or

$$n + 1 = p, \qquad n + 2 = 2^b,$$

with b prime.

The two infinite families arising before the Zsigmondy elimination are

$$n = 2^{2^t} - 1$$

and

$$n = 2^b - 2, \qquad b \equiv 1 \pmod 4.$$

The latter reduces further to the Fermat-prime-shaped family

$$b = 2^{2^s} + 1.$$

Branch $d_2 = 2, d_3 = 1, c > 1$

$$n = 3^p - 3,$$
$$\frac{3^p - 1}{2} = \Phi_p(3),$$
$$\frac{3^p + 1}{4} = \Phi_{2p}(3),$$
$$3^p - 2 = P^\alpha.$$

The two decisive congruence regimes are

$$p \equiv 1 \pmod 4$$

and

$$p \equiv 3 \pmod 4.$$

In the first,

$$p - 1 = 2^t$$

is forced by the prime-power condition at $k = 6$, followed by

$$13 \mid 3^p + 4$$

at $k = 7$.

In the second,

$$5 \mid 3^p - 2,$$

hence

$$3^p - 2 = 5^{2a},$$

reducing to

$$(5^a)^2 + 2 = 3^p.$$

Branch $d_2 = 2, d_3 = 3$

$$12 \mid n,$$
$$d_4 \in \{1, 4\},$$
$$d_4 = 1 \Longrightarrow n = 12,$$

and for $n > 12$,

$$d_4 = 4, \qquad d_5 \in \{1, 5\}.$$

For $d_5 = 5$,

$$n = 120u$$

and

$$2q_2 - q_1 = 1,$$
$$3q_3 - 2q_2 = 1,$$
$$4q_4 - 3q_3 = 1,$$
$$5q_5 - 4q_4 = 1.$$

If $d_6 = 2$,

$$20u + 1 = 3^{4t}, \qquad u = \frac{3^{4t} - 1}{20}.$$

C A checklist for the next proof phase

The natural next step is not a larger blind search. It is to attack the remaining leaves one by one.

1. Study the equation
$$P^\alpha + 1 = 2Q^\beta$$
together with the primality of
$$P^\alpha + 2.$$
The known values $n = 8, 16, 80$ should emerge naturally.

2. Study the four consecutive relations
$$p - 4 = P^\alpha, \quad p - 3 = 2Q^\beta, \quad p - 2 = 3R^\gamma, \quad p - 1 = 4S^\delta.$$
The first relevant example in this branch is $p = 53$ (equivalently $n = 48$).

3. For $d_5 = 5$, exploit the linear prime-power chain and try to force incompatible residue classes.

4. In the $d_6 = 2$ exponential branch, apply primitive-divisor arguments to the numbers
$$3^{4t+1} - 2, \quad 2 \cdot 3^{4t} - 1, \quad \frac{3^{4t+1} - 1}{2}, \quad \frac{6 \cdot 3^{4t} - 1}{5}.$$

5. Extend the d_k-classification to $k = 7, 8, \ldots$ only after the small-k leaves have been simplified as far as possible.